

Here, $\text{End}_{\mathbb{F}_p}$ refers to the $\mathbb{F}_p$ -algebra of $\mathbb{F}_p$ -module endomorphisms with $\circ$ as the multiplication. Here, α is well-defined since d commutes with each λ_u . Applying α to eq. (9) yields

$$d^{p^m} - d = \prod_{u \in \mathbb{F}_p[a]} (d - \lambda_u)$$

in $\text{End}_{\mathbb{F}_p}(A)$. The left side vanishes because of eq. (8). Therefore, we arrive at

$$0 = \prod_{0 \neq u \in \mathbb{F}_p[a]} (d - \lambda_u) \circ d. \quad (10)$$

We claim that $d - \lambda_u : A \rightarrow A$ is injective for all $0 \neq u \in \mathbb{F}_p[a]$. In fact, if $(d - \lambda_u)(b) = 0$, then $ab - ba = ub$, so that $ba = (a - u)b$. Since $a - u \in \mathbb{F}_p[a]$, there is a polynomial $f \in \mathbb{F}_p[T]$ of degree $< m \leq k$ with $a - u = f(a)$. Thus, $ba = f(a)b$. Since the constructive Wedderburn Theorem $W_{p,k,f}$ holds by assumption, it follows $ab = ba$, which means $ub = 0$. Since $u \neq 0$ and $\mathbb{F}_p[a]$ is a field, u is a unit, so that $b = 0$, and we are done.

Since all $d - \lambda_u$ ($u \neq 0$) are injective, eq. (10) implies $d = 0$, and a is central. $\square$

The next step is to transform the proof into an equational proof.

Theorem 7.5. *Let $p \in \mathbb{P}$ and $k \geq 1$. Assume that for all $f \in \mathbb{F}_p[T]$ of degree $< k$ there is an equational proof of the constructive Wedderburn Theorem $W_{p,k,f}$. Then there is an equational proof that every p^k -ring with $p = 0$ is commutative.*

Proof. The first step is to describe a product decomposition $\mathbb{F}_p[T]/\langle T^{p^k} - T \rangle \cong \prod_i \mathbb{F}_{p^{m_i}}$ in more explicit terms. Let S be the finite set of all monic irreducible polynomials over $\mathbb{F}_p$ whose degree divides k . This can be computed with standard methods. For example, when $p = 2$ and $k = 2$ we have $S = \{T, T + 1, T^2 + T + 1\}$. Let $g \in S$. Then g and $\prod_{g' \in S \setminus \{g\}} g'$ are coprime, so that the extended Euclidean algorithm yields polynomials u, v with

$$u \cdot g + v \cdot \prod_{g' \in S \setminus \{g\}} g' = 1.$$

Define $e_g := v \cdot \prod_{g' \in S \setminus \{g\}} g'$. Then $e_g \equiv 1 \pmod{g}$ and $e_g \equiv 0 \pmod{g'}$ for all $g' \in S \setminus \{g\}$. We have

$$T^{p^k} - T = \prod_{g \in S} g. \quad (11)$$

The abstract reason is that both sides equal the product of all $T - u$ with $u \in \mathbb{F}_{p^k}$, but in practice, for fixed p and k we can just verify this by expanding the product. It follows that the images $[e_g] \in \mathbb{F}_p[T]/\langle T^{p^k} - T \rangle$ are idempotent, pairwise orthogonal³, and their sum is 1.

Let A be a p^k -ring with $p = 0$, and let $a \in A$. We want to show that a is central. There is a homomorphism of $\mathbb{F}_p$ -algebras

$$\mathbb{F}_p[T]/\langle T^{p^k} - T \rangle \rightarrow A$$

mapping $[T]$ to a , and thus $[e_g]$ to $e_g(a)$. It follows that all the elements $e_g(a) \in A$ are idempotent, pairwise orthogonal, and their sum is 1, meaning

$$e_g(a)^2 = e_g(a), \quad e_g(a) \cdot e_{g'}(a) = 0 \ (g \neq g'), \quad \sum_{g \in S} e_g(a) = 1. \quad (12)$$

For an equational proof of these equations in A , we cannot use homomorphisms and quotients of polynomial rings, but we can just verify them by hand, only using the relation $a^{p^k} = a$. In order to show that $a = \sum_{g \in S} a \cdot e_g(a)$ is central, it is then enough to show that each $a \cdot e_g(a)$ commutes with each $b \cdot e_g(a)$, where $b \in A$ and $g \in S$. Both elements stay the same when multiplied with $e_g(a)$, which is central by Lemma 2.8. Hence, we may assume that $e_g(a)$ is the multiplicative identity of our ring. In fact, we just have to replace every occurrence of $1 \in A$ below by $e_g(a)$, and the calculations remain valid. Of course, this corresponds to the product decomposition in our proof of Theorem 7.4, but what we do here is an equational version of this argument.

³Two idempotents e, f are called orthogonal if $ef = 0$.

Strictly speaking, we are not allowed to speak of “irreducible polynomials” in an equational proof. The construction above, however, just explains the *method* how to write down an equational proof for a fixed pair (p, k) , in which we are just using the polynomial expressions $e_g(a)$ and $g(a)$ in A . This will become more clear in the examples below.

So, we have a polynomial $g \in S$ with $e_g(a) = 1$, and this implies $e_{g'}(a) = 0$ for $g' \neq g$. Next, we claim that

$$g(a) = 0 \tag{13}$$

holds in A . To see this, notice that for all $g' \neq g$ we have $1 = e_g(a) \equiv 0 \pmod{g'(a)}$, i.e. $g'(a)$ is a unit. On the other hand, by eq. (11) we have $0 = a^{p^k} - a = \prod_{g' \in S} g'(a)$. Since all factors $g'(a)$ with $g' \neq g$ are units, we must have $g(a) = 0$. Conversely, $g(a) = 0$ implies $e_g(a) = 1$ since $e_g \equiv 1 \pmod{g}$.

The whole preceding discussion was just about finding some $g \in S$ with $g(a) = 0$. This is the only thing we will need from it in the following. In classical logic, this means that $\mathbb{F}_p[a]$ is either zero or a field.

Let $m := \deg(g)$. If $m = 1$, then $a = u \cdot 1$ for some $u \in \mathbb{F}_p$ and we are done. So assume $m > 1$ (and hence $k > 1$). Notice that we are *not* using the LEM here. What we are writing down is one proof for every $g \in S$. When $S = \{T, T+1, \dots\}$, we start with $g = T$, then $g = T+1$, etc. But for linear polynomials the proof is easy. So we switch directly to the next polynomials. Incidentally, this means that the proof is easy for $k = 1$, namely exactly the one in Theorem 5.5.

Since g is irreducible of degree > 1 , the constant term of g is a unit in $\mathbb{F}_p$. So clearly, $a \in A$ is a unit. Since g is irreducible of degree m , we have $g \mid T^{p^m} - T$ (this can be verified manually with a polynomial division), so that $a^{p^m} = a$ by eq. (13).

Let $b \in A$. We want to show that a and b commute. Choose an enumeration $f_0, f_1, \dots, f_{p^m-1}$ of the set $\mathbb{F}_p[T]_{<m}$ of polynomials over $\mathbb{F}_p$ of degree $< m$, starting with $f_0 = 0$. We define a sequence of elements $b_0, b_1, b_2, \dots, b_{p^m}$ in A recursively by

$$b_0 := b, \quad b_{n+1} := ab_n - b_na - f_n(a)b_n. \tag{14}$$

For example, $b_1 = ab - ba$, and the goal is to prove $b_1 = 0$. We claim that

$$b_{p^m} = 0. \tag{15}$$

Unfortunately, our proof is not equational, but it is constructive, and eq. (15) can be verified in every example of p, k, g manually as well, just by using $g(a) = 0$ (see the examples below). Define the map $d : A \rightarrow A$ by $d(x) := ax - xa$. As before, we prove $d^{p^m} = d$, only using $a^{p^m} = a$. The definition of b_n becomes

$$b_n = ((d - f_{n-1}(a)) \circ \dots \circ (d - f_0(a)))(b).$$

Hence,

$$b_{p^m} = \prod_{f \in \mathbb{F}_p[T]_{<m}} (d - f(a))(b),$$

so that it suffices to prove the equation

$$T^{p^m} - T = \prod_{f \in \mathbb{F}_p[X]_{<m}} (T - f(a))$$

in $A[T]$. Since $g(a) = 0$, there is a homomorphism of $\mathbb{F}_p$ -algebras $\mathbb{F}_p[X]/\langle g(X) \rangle \rightarrow A$, $[X] \mapsto a$, and it suffices to prove

$$T^{p^m} - T = \prod_{f \in \mathbb{F}_p[X]_{<m}} (T - [f])$$

in $(\mathbb{F}_p[X]/\langle g(X) \rangle)[T]$. This follows from the fact that $\mathbb{F}_p[X]/\langle g(X) \rangle$ is a field with p^m elements, namely those $[f]$ with $f \in \mathbb{F}_p[X]_{<m}$. This finishes the proof of eq. (15).

Next, we claim that for $n \geq 1$ there is an equational proof of

$$b_{n+1} = 0 \implies b_n = 0. \tag{16}$$

If $b_{n+1} = 0$, this means $b_n a = (a - f_n(a))b_n = f'(a)b_n$ with $f' := T - f_n$. Notice that f' has degree $< k$ since $k > 1$ and f_n has degree $< m \leq k$. By assumption, there is an equational proof of the constructive Wedderburn Theorem $W_{p,k,f'}$, so that $b_n a = a b_n$, meaning $f_n(a)b_n = 0$. We claim that $f_n(a)$ is a unit in A , which leads to $b_n = 0$. In fact, f_n and g are coprime (since $f_n \neq 0$ has degree $< m = \deg(g)$ and g is irreducible), so that $u f_n + v g = 1$ for some computable polynomials u, v . Then $1 = u(a)f_n(a) + v(a)g(a) = u(a)f_n(a)$, proving our claim.

Combining eq. (15) and eq. (16) gives $b_1 = 0$, so that a, b commute. $\square$

Remark 7.6. If two polynomials $g, g' \in S$ only differ by a linear substitution, then the proof for g in Theorem 7.5 (that $g(a) = 0$ implies that a is central) leads to a proof for g' , so that g' may be omitted. In fact, if $g' = w g(uT + v)$ for $v \in \mathbb{F}_p$ and $u, w \in \mathbb{F}_p^\times$, then the assumption $g'(a) = 0$ means $g(ua + v) = 0$. If there is a proof for g , then $ua + v$ is central and hence a too. Similarly, if g, g' are reciprocal polynomials (up to a unit), then the proof for g leads to a proof for g' . In fact, if $g'(a) = 0$, then a is a unit with $a^m g(a^{-1}) = 0$, so that $g(a^{-1}) = 0$. If there is a proof for g , then a^{-1} is central, so that a is central as well. Again, one can write some SageMath code that computes a sufficient set of polynomials for us.

We will now show how the equational proof provided by Theorem 7.5 looks like in the example $p = 2$ and $k = 2$. It will be considerably more complicated than ours in Theorem 6.4, but it explains the method (that works for *all* $k \geq 1$) quite well.

Example 7.7. Consider $p = 2$ and $k = 2$. So we will show (again) that every 4-ring A is commutative. We have $S = \{T, T + 1, T^2 + T + 1\}$. The extended Euclidean algorithm yields the equations

$$\begin{aligned} T^2 \cdot (T) + 1 \cdot (T + 1)(T^2 + T + 1) &= 1 \\ (T^2 + 1) \cdot (T + 1) + 1 \cdot T(T^2 + T + 1) &= 1 \\ 1 \cdot (T^2 + T + 1) + 1 \cdot T(T + 1) &= 1 \end{aligned}$$

Therefore, we define

$$\begin{aligned} e_T &:= (T + 1)(T^2 + T + 1) = T^3 + 1, \\ e_{T+1} &:= T(T^2 + T + 1) = T^3 + T^2 + T, \\ e_{T^2+T+1} &:= T(T + 1) = T^2 + T. \end{aligned}$$

Let $a \in A$. Then the elements $e_T(a)$, $e_{T+1}(a)$, $e_{T^2+T+1}(a)$ are idempotent (hence central), pairwise orthogonal, and their sum is 1. We can verify this by a direct calculation, just using $a^4 = a$ (and of course $2 = 0$ in A). We only show two examples: $a^3 + 1$ is idempotent since $(a^3 + 1)^2 = a^6 + 1 = a^3 + 1$, and $(a^3 + 1)(a^2 + a) = (a^4 + a)(a + 1) = 0(a + 1) = 0$. It suffices to prove that every $a \cdot e_g(a)$ commutes with all $b \cdot e_g(a)$, where $g \in S$ and $b \in A$. For $g = T$ we find

$$a \cdot e_T(a) = a \cdot (a^3 + 1) = a^4 + a = 0,$$

so this case is trivial. For $g = T + 1$ we find

$$a \cdot e_{T+1}(a) = a^4 + a^3 + a^2 = e_{T+1}(a),$$

so this case is also trivial. Linear polynomials will always be trivial. The only interesting case is $g = T^2 + T + 1$. Here,

$$a \cdot e_g(a) = a^3 + a^2.$$

Now, we do not necessarily have $g(a) = a^2 + a + 1 = 0$ as in eq. (13), but here is why we may assume it: A direct calculation shows

$$(a^3 + a^2)^2 + (a^3 + a^2) + (a^2 + a) = 0.$$

So $c := a \cdot e_g(a)$ satisfies $c^2 + c + 1 = 0$ in the 4-ring $e_g(a)A$ with multiplicative identity $e_g(a)$, and it suffices to prove that c is central in $e_g(a)A$. Formally, this ring does not even exist in